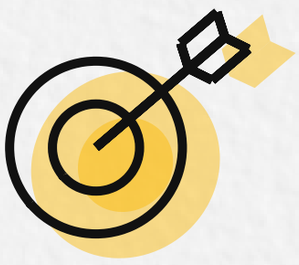
Week 1- Learning Map



Goal: Understand how Agentic AI systems work by learning the core concepts of LLMs, agents, tools, and reasoning loops.

By the end of the Week 1 session, you will be able to build a simple agent workflow using LLMs and tool calls.

Overview...

In Week 1, we will introduce the fundamentals of Agentic AI and how it differs from prompt-based systems. You will learn how agents use reasoning loops, tools, and routing to solve complex tasks.

We will walk through how to interact with LLMs using APIs and SDKs, and understand how tool calling enables agents to perform real-world actions.

The session will also cover the structure of an agent loop (think → act → observe) and how it is used to build intelligent systems.

You will also learn the Routing pattern, where the agent decides the next step based on input, context, or predefined logic.

Through hands-on examples, you will implement a basic agent with tool calling and routing, helping you connect concepts to practical use cases.

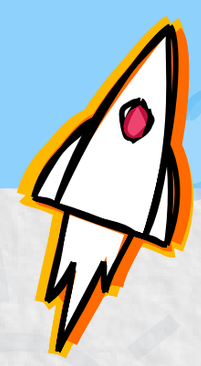
We will be building a CRM Lead Qualification Agent project to apply these concepts in a real-world scenario.

By the end of the class, you will have a clear understanding of agent architecture and how to design agent workflows.

Note:

Pre-class and post-class resources are optional, but the pre-class setup must be completed before the live class.

HOW TO START?



Roadmap

Pre-class resources (Optional)

1) Prompting
2) Introduction to Agents
3) Client Libraries primer

Mandatory setup for Week 1

- OpenAI API Key
- Google Colab Setup

Week 1 Live Class

- Reflex & Goal-based agents
- Utility agents and Intro to LLMs
- LLM-Based Agents
- CRM Lead Qualification Agent
- Week 0 Activity Discussion

Post-class resources (Optional deep dives)

- LLMs: overview → limits → hallucinations
- Transformers (bird's eye) → RAG motivation
- Agents deep dive + LangChain + Func calls

Assignment

- Build an AI agent that acts as the "first responder" for server incidents: Investigate, Act & Escalate

Please use the Wednesday Technical Coaching session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Thursday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.